What Can Existing Logs Establish About a Change?

Log of a gate that runs every call → replay the change κ → three cases

Replay a fixed change κ

Agree



Agrees throughout

Outcome law under the change: **Read from the log**

Stop

First deviation

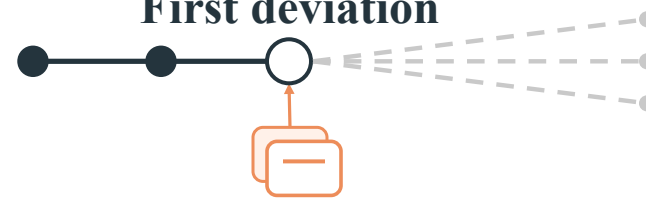


Needs per-step harm in the log

Outcome law under the change: **Declared stop value**

Continue

First deviation



Restore; run only the continuation

Outcome law under the change: **Support, snapshot run, or bound**

$$P^{\kappa}(L \in B \mid r) = P^{0}(L \in B, \text{agree} \mid r) + E^{0}[b_{\kappa}(B \mid h_{\kappa}^{+})\,\mathbf{1}\{\text{stop}\} \mid r] + P^{\kappa}(L \in B, \text{continue} \mid r)$$